EVENT OVERVIEW

Select one earthquake. Counts use the current building stock and 2023 population (as-if-today).

EVENT

Event

Kaikōura, 2016

SOURCE AND BASIS

Event	Shaking Source	Exposure Basis
Kaikōura, 2016	official	As-if-today

MMI THRESHOLD

MMI Threshold

6

PEAK MAGNITUDE

7.8

Magnitude

MAXIMUM MMI

8.4

Max MMI

EXPOSED BUILDINGS

78,564

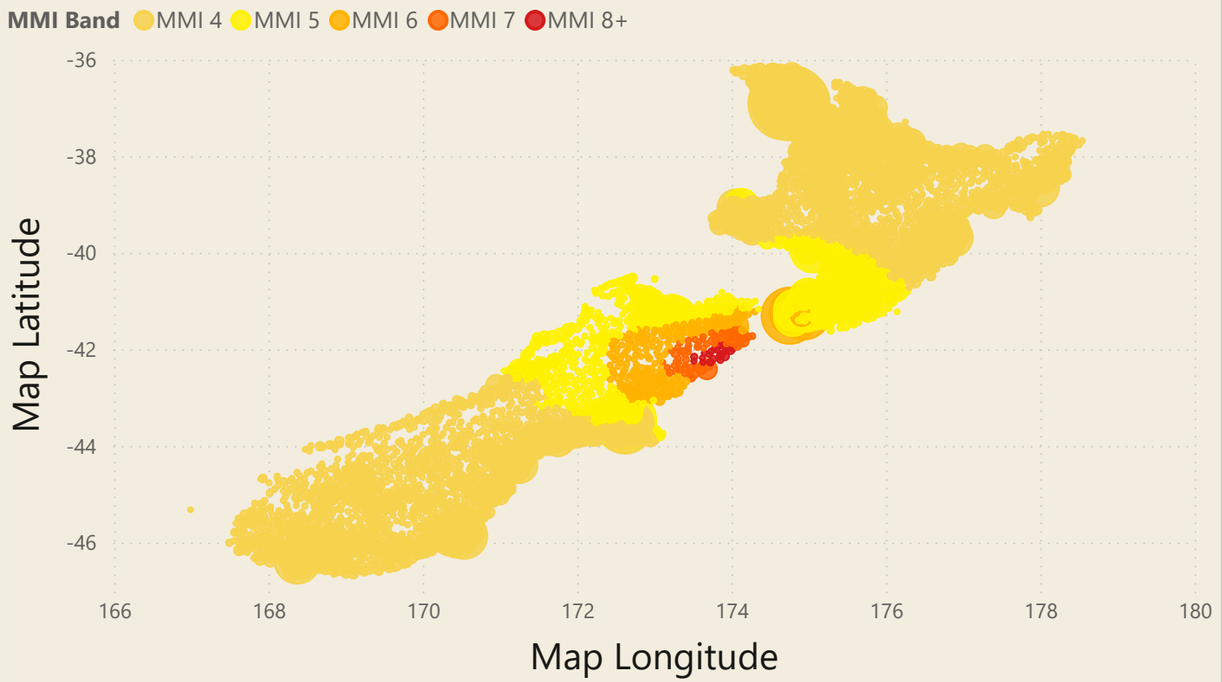
Exposed Buildings

EXPOSED PEOPLE

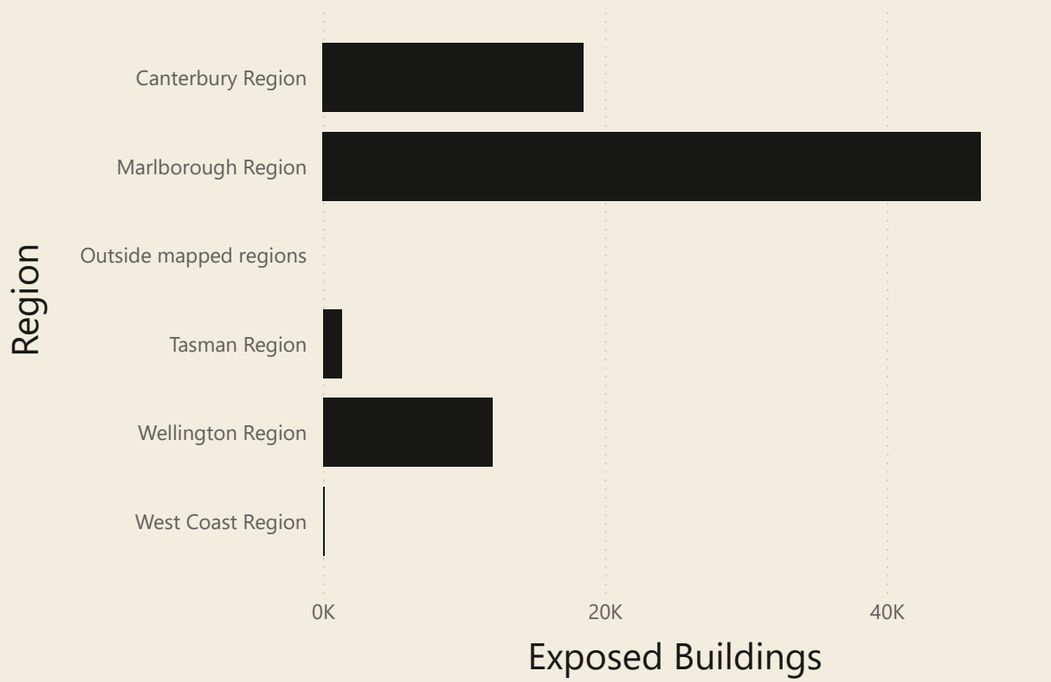
83,276

Exposed Population

H3 EXPOSURE POINTS / LONGITUDE × LATITUDE



REGIONS BY EXPOSED BUILDINGS



NZ HAZARD EXPOSURE MONITOR / AS-IF-TODAY REPLAY / 2023 POPULATION

EVENT COMPARISON

Ten fixed earthquakes replayed against today's buildings and the 2023 population; totals count repeat exposure.

HISTORICAL EVENTS

10

Event Count

EVENT-BUILDING EXPOSURES

1,031,371

Exposed Buildings

EVENT-PEOPLE EXPOSURES

1,621,964

Exposed Population

HIGHEST MMI

9.4

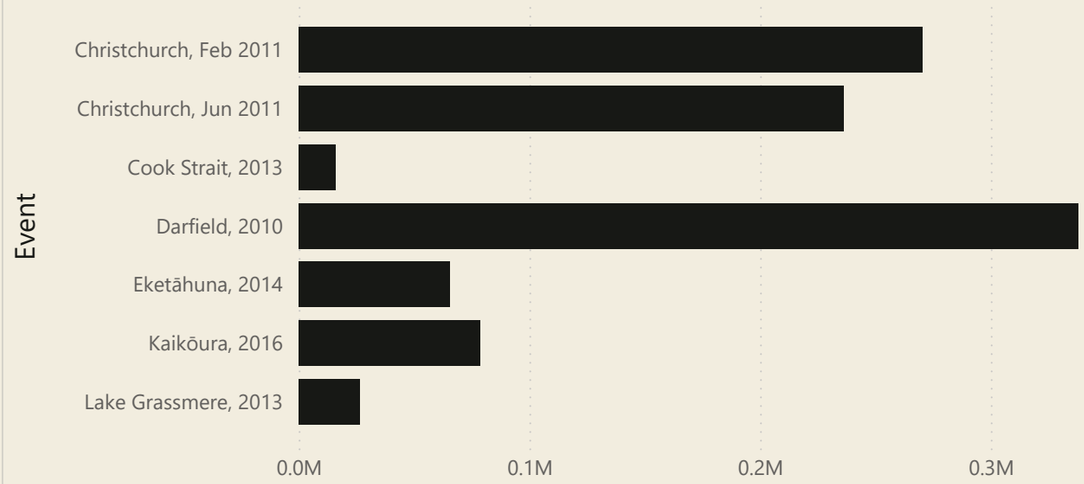
Max MMI

MMI THRESHOLD

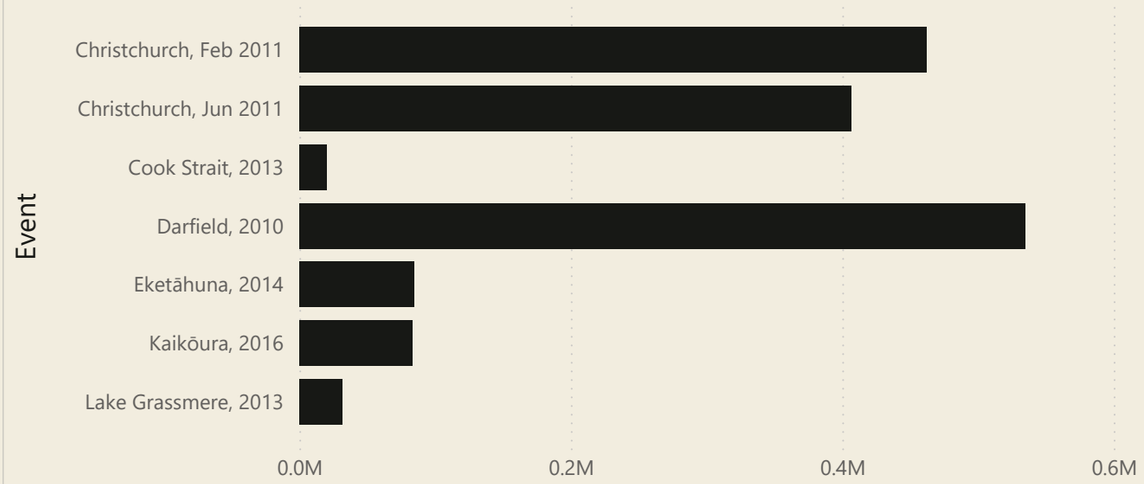
MMI Threshold

6

BUILDINGS / EVENT



PEOPLE / EVENT



EVENT COMPARISON / SELECT ANY ROW TO CROSS-FILTER

Event	Magnitude	Max MMI	Exposed Buildings	Exposed Population
Christchurch, Feb 2011	6.2	9.4	270,518	461,765
Christchurch, Jun 2011	6.0	8.8	236,306	406,499
Cook Strait, 2013	6.5	7.6	16,059	19,950
Total	**7.9**	**9.4**	**1,031,371**	**1,621,964**

REGIONAL EXPOSURE

Cell-to-region attribution follows the majority building location; boundaries are approximate.

EVENT

Event

Kaikōura, 2016

MMI THRESHOLD

MMI Threshold

6

EXPOSED BUILDINGS

78,564

Exposed Buildings

SHARE OF REGION STOCK

2.4%

% Region Buildings Exposed

ACCUMULATED EXPOSURES

1,031,371

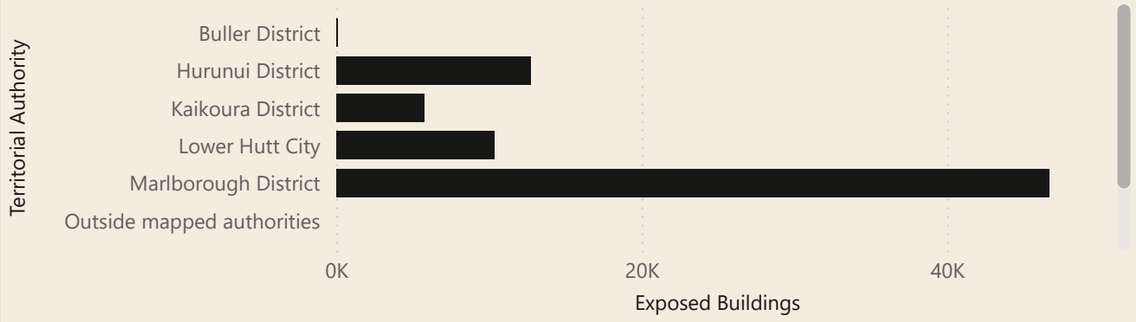
Accumulated Building Exposures

BUILDINGS IN MMI 6+ CELLS

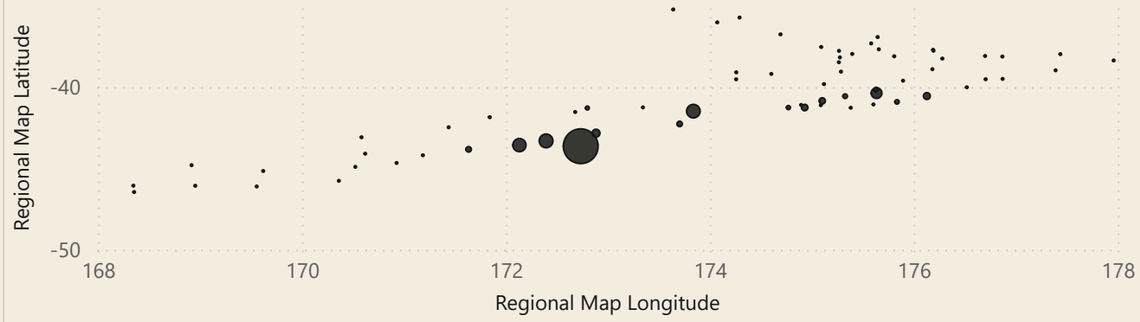
491,689

Buildings in Strong Cells

REGION / TERRITORIAL AUTHORITY RANKING



ACCUMULATION CENTROIDS / LONGITUDE × LATITUDE



REGIONAL DETAIL

Region	Exposed Buildings	% Region Buildings Exposed	Accumulated Building Exposures
⊞ Auckland Region			0
⊞ Bay of Plenty Region			0
⊞ Canterbury Region	18,496	3.9%	863,320
⊞ Gisborne Region			0
⊞ Hawke's Bay Region			0
Total	**78,564**	**2.4%**	**1,031,371**

NZ HAZARD EXPOSURE MONITOR / AS-IF-TODAY REPLAY / 2023 POPULATION

EVENT HISTORY

Event time along the horizontal axis, magnitude vertically, bubble size by thresholded building exposure.

REGION

Region

- [ ] Auckland Region

MMI THRESHOLD

MMI Threshold

6

EVENTS

10

Event Count

EVENT-BUILDING EXPOSURES

1,031,371

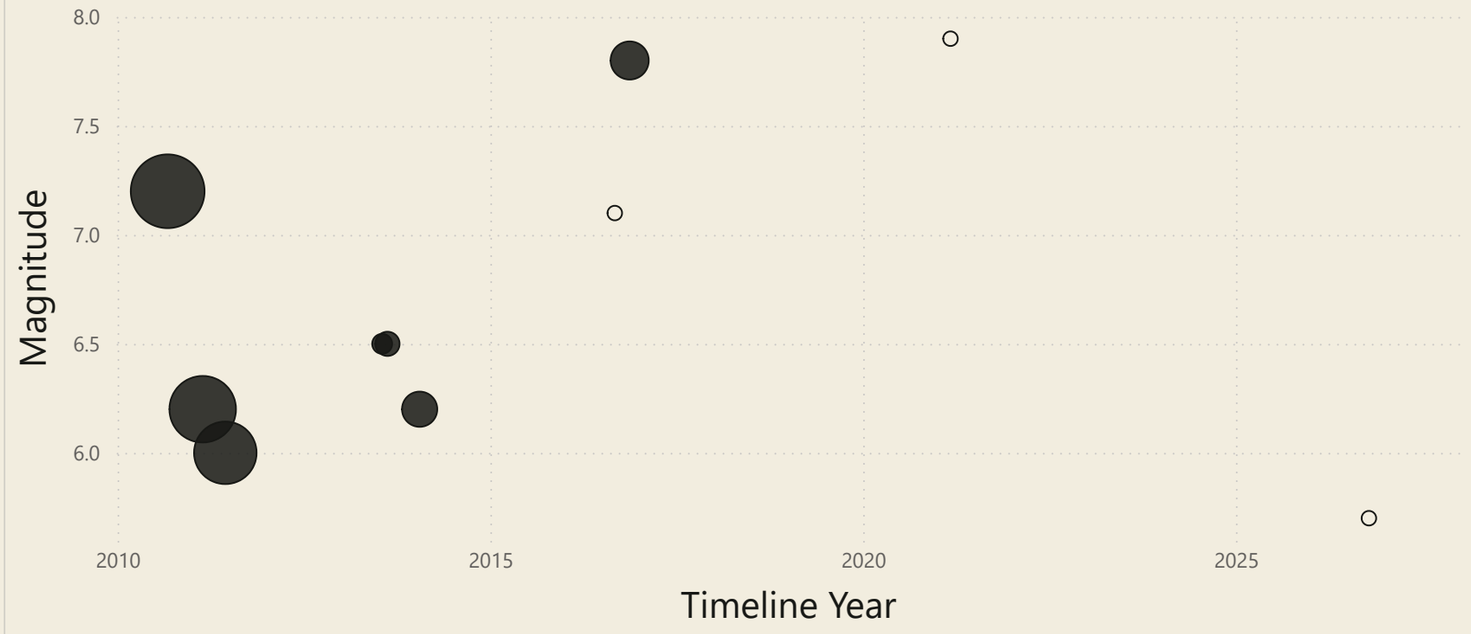
Exposed Buildings

HIGHEST MMI

9.4

Max MMI

EVENT TIME × MAGNITUDE / BUBBLE AREA = EXPOSED BUILDINGS



EVENT CHRONOLOGY

Event	Event Date (NZ)	Magnitude	Exposed Buildings
Christchurch, Feb 2011	22 Feb 2011	6.2	270,518
Christchurch, Jun 2011	13 Jun 2011	6.0	236,306
Cook Strait, 2013	21 Jul 2013	6.5	16,059
Darfield, 2010	4 Sep 2010	7.2	337,971
East Cape, 2016	2 Sep 2016	7.1	
Eketāhuna, 2014	20 Jan 2014	6.2	65,461
Kaikōura, 2016	14 Nov 2016	7.8	78,564
Kermadec Islands, 2021	5 Mar 2021	7.9	
Lake Grassmere, 2013	16 Aug 2013	6.5	26,492
Taumarunui, Oct 2026	15 Oct 2026	5.7	
Total		**7.9**	**1,031,371**

DATA HEALTH

Local gold and SQLite snapshot; this page does not claim live Azure health.

LAST LOCAL RUN

29 Oct 2026 00:20

Last Successful Run

AGE OF LOCAL RUN / HOURS

167

Hours Since Last Run

RECORDED FAILURES

0

Failed Runs

VALID STATIONS

2,547

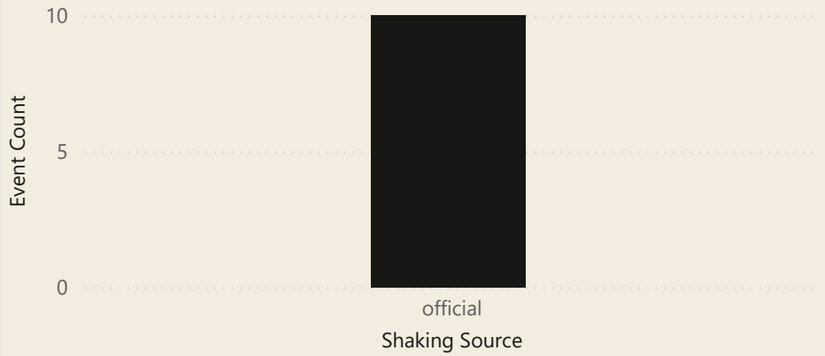
Validation Stations

STATION-WEIGHTED MAE

1.049

Estimated MMI MAE

SHAKING SOURCE / EVENT COUNT



MODEL VALIDATION BY EVENT

Event	Validation Stations	Estimated MMI MAE
Christchurch, Feb 2011	201	0.861
Christchurch, Jun 2011	193	0.849
Cook Strait, 2013	360	0.852
Darfield, 2010	425	1.160
East Cape, 2016	151	1.136
Eketāhuna, 2014	423	0.641
Kaikōura, 2016	323	1.373
Kermadec Islands, 2021	1	2.821
Lake Grassmere, 2013	289	0.787
Taumarunui, Oct 2026	181	2.309
Total	**2,547**	**1.049**

LOCAL SNAPSHOT ONLY. No policy or insured-value data are used. MAE is station-weighted and indicative, not an actuarial loss estimate. Building lifecycle dates are approximate; all ten historical replays use current outlines. Water cells and the Chatham Islands are outside the mapped exposure model.